

Nolan McMahon

215-692-3848 | mcmahonn3@wit.edu | linkedin.com/in/nolanryanmcmahon | Brookline, MA

TECHNICAL SKILLS

Software & Tools: SolidWorks, ANSYS Fluent, Revit, AutoCAD MEP, Fusion 360, Onshape, MATLAB, Python (NumPy, SciPy), Procore, Microsoft Office Suite

Embedded & DAQ: Arduino (C++), ESP32, ADC Signal Conditioning, Serial Data Logging

Engineering Knowledge: Heat Transfer, Fluid Mechanics, Mechanical Vibrations, FEA, Machine Element Design, GD&T, Tolerance Analysis, Technical Documentation

Codes & Standards: IBC, IMC, IECC

EDUCATION

B.S. Mechanical Engineering | Wentworth Institute of Technology, Boston, MA *Expected August 2026*

Recent Awards: Dean's List, Spring 2026 | U.S. Patent Pending | Relevant Coursework: Mechanical Vibrations, FEA & Simulation-Based Design, Heat Transfer, Machine Element Design, Additive Manufacturing

EXPERIENCE

Mechanical Forensics / Code Advisory Engineer Intern | Rimkus Consulting Group, Boston, MA *Sep – Dec 2025*

- Reconstructed vehicle collision sequences across 4 cases — including an electric vehicle and multiple tractor-trailers — by pulling and interpreting airbag control module data and onboard computer records to establish fault, impact sequence, and damage causation.
- Investigated mechanical failure events including pipe thread bursts and a vintage industrial fan injury case; performed metallographic analysis and produced evidence-based conclusions on failure mode, material condition, and applicable code compliance.
- Prepared third-party plan review packages for the DC Department of Buildings, including code compliance matrices covering IBC, IMC, IECC, and NFPA 241 — work that directly supported project approvals for a licensed engineering firm.
- Drafted mechanical and plumbing layouts in AutoCAD MEP and Revit; identified gaps in existing QA workflows and built improved documentation tools that engineers on the team continued using after the co-op ended.

Commissioning Intern | Integra Mission Critical, Austin, TX *Jan – Apr 2025*

- Worked on commissioning of a live 80 MW data center, catching 2–3 system-level issues per day — including power cables routed in ways that created dangerous heat loads and vents left unsecured ahead of generator startups — before they became costly problems.
- Managed 400+ commissioning tracking logs across 4 facility floors, covering power distribution, medium-voltage step-downs, diesel generators, air-cooled chillers, and fan walls.
- Referenced technical drawings and manufacturer specifications for joint pack installations to verify contractor work in the field; flagged non-conforming installations before energization and applied the same review process across multiple systems and devices throughout the facility.
- Maintained arc flash postings across active commissioning zones and kept commissioning schedules on track across multiple subcontractors and client stakeholders simultaneously.

ENGINEERING PROJECTS

Server Thermal Diagnostic System | *Independent Project* | **U.S. Patent Pending**

- Identified a gap in how thermal issues are caught in server and PC cooling systems and designed a multi-channel diagnostic tool to detect thermal gradients, airflow disruptions, and heat-transfer irregularities that standard monitoring misses.
- Built and tested a working prototype using Arduino-based data acquisition, MATLAB signal analysis, and SolidWorks thermal modeling; applied FEA to validate mounting geometry and thermal stress behavior under load.
- Plans to validate cooling-path behavior further using ANSYS Fluent CFD analysis on test setup 2 to characterize airflow and thermal performance under controlled conditions.
- Concept was filed for patent protection and has since drawn genuine interest from engineers and faculty who called it a novel and practical approach to a real problem in electronics cooling.
- U.S. Provisional Patent Application No. 64/086,631 — filed June 2026.

Myoelectric Prosthetic Hand System | **Senior Capstone**

- Designing a prosthetic hand system built to fit a wide range of users — ball bearing guided compression platform adjusters accommodate different hand sizes without custom fabrication, targeting a sub-500g form factor and sub-\$500 material cost to make the device viable for users who can't access traditional prosthetic pricing.
- Designed palm housing and 3-link finger assembly in SolidWorks; developed a tendon-driven kinematic model using Jacobian matrix formulation to accurately relate motor torque to fingertip force across a 3-DOF system.
- Integrated EMG signal acquisition, precision motor drive electronics, and Li-Po power management into a unified wearable system; design developed in consideration of ISO 13485 (medical device quality management) and ISO 10328 (structural testing for prosthetic components).

INTERESTS & ACTIVITIES

Brookline Masters Swimming | 3D Printing & Prototyping | Vintage Baseball Card Collecting & Trading | Strength Training